The intervals are the pigeonholes and the d_j's are the pigeons. If the interval $(0, 1/n)$ or $(-1/n, 0)$ is occupied, then we are done, since the d_j in that interval tells us which j makes $|d_j| < 1/n$. If not, then there are n pigeons for at most $n-2$ pigeonholes, so by the pigeonhole principle there is some interval, say $\big((k-1)/n, k/n\big)$, with two pigeons in it, say d_r and d_s, with $r < s$. Now we will consider $sx - rx$ and show that it is within $1/n$ of its nearest integer; that will complete our proof, since $sx - rx = (s-r)x$, and $s-r$ is a positive integer less than n.

We can write $rx = N(rx) + d_r$ and $sx = N(sx) + d_s$, where $(k-1)/n \le d_r < k/n$ and $(k-1)/n \le d_s < k/n$. Subtracting, we have that $sx - rx = [N(sx) - N(rx)] + [d_s - d_r]$. Now the quantity in the first pair of square brackets is an integer. Furthermore the quantity in the second pair of square brackets is the difference of two numbers in the interval $\big((k-1)/n, k/n\big)$ and hence has absolute value less than $1/n$ (the extreme case would be when one of them is very close to $(k-1)/n$ and the other is very close to k/n). Therefore, by definition of "closest integer" $sx - rx$ is at most a distance $1/n$ from its closest integer, i.e., $|(sx - rx) - N(sx - rx)| < 1/n$, as desired. (The case in which n is odd is similar, but we need to extend our intervals slightly past $\pm 1/2$, using $n+1$ intervals rather than n. This is okay, since when we subtract 2 from $n+1$ we still have more pigeons than pigeonholes.)

47. **a)** Assuming that each $i_k \le n$, there are only n pigeonholes (namely $1, 2, \ldots, n$) for the n^2+1 numbers $i_1, i_2, \ldots, i_{n^2+1}$. Hence, by the generalized pigeonhole principle at least $\lceil (n^2+1)/n \rceil = n+1$ of the numbers are in the same pigeonhole, i.e., equal.

b) If $a_{k_j} < a_{k_{j+1}}$, then the subsequence consisting of a_{k_j} followed by a maximal increasing subsequence of length $i_{k_{j+1}}$ starting at $a_{k_{j+1}}$ contradicts the fact that $i_{k_j} = i_{k_{j+1}}$. Hence $a_{k_j} > a_{k_{j+1}}$.

c) If there is no increasing subsequence of length greater than n, then parts **(a)** and **(b)** apply. Therefore we have $a_{k_{n+1}} > a_{k_n} > \cdots > a_{k_2} > a_{k_1}$, a decreasing subsequence of length $n+1$.

SECTION 6.3 Permutations and Combinations

In this section we look at counting problems more systematically than in Section 6.1. We have some formulae that apply in many instances, and the trick is to recognize the instances. If an ordered arrangement without repetitions is asked for, then the formula for permutations usually applies; if an unordered selection without repetition is asked for, then the formula for combinations usually applies. Of course the product rule and the sum rule (and common sense and cleverness) are still needed to solve some of these problems—having formulae for permutations and combinations definitely does not reduce the solving of counting problems to a mechanical algorithm.

Again the general comments of Section 6.1 apply. Try to solve problems more than one way and come up with the same answer—you will learn from the process of looking at the same problem from two or more different angles, and you will be (almost) sure that your answer is correct.

1. Permutations are ordered arrangements. Thus we need to list all the ordered arrangements of all 3 of these letters. There are 6 such: a, b, c; a, c, b; b, a, c; b, c, a; c, a, b; and c, b, a. Note that we have listed them in alphabetical order. Algorithms for generating permutations and combinations are discussed in Section 6.6.

3. If we want the permutation to end with a, then we may as well forget about the a, and just count the number of permutations of $\{b, c, d, e, f, g\}$. Each permutation of these 6 letters, followed by a, will be a permutation of the desired type, and conversely. Therefore the answer is $P(6, 6) = 6! = 720$.

5. We simply plug into the formula $P(n,r) = n(n-1)(n-2)\cdots(n-r+1)$, given in Theorem 1. Note that there are r terms in this product, starting with n. This is the same as $P(n,r) = n!/(n-r)!$, but the latter formula is not as nice for computation, since it ignores the fact that each of the factors in the denominator cancels one factor in the numerator. Thus to compute $n!$ and $(n-r)!$ and then to divide is to do a lot of extra arithmetic. Of course if the denominator is 1, then there is no extra work, so we note that $P(n,n) = P(n,n-1) = n!$.
a) $P(6,3) = 6 \cdot 5 \cdot 4 = 120$ **b)** $P(6,5) = 6! = 720$ **c)** $P(8,1) = 8$
d) $P(8,5) = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 6720$ **e)** $P(8,8) = 8! = 40{,}320$ **f)** $P(10,9) = 10! = 3{,}628{,}800$

7. This is $P(9,5) = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 15{,}120$ by Theorem 1.

9. We need to pick 3 horses from the 12 horses in the race, and we need to arrange them in order (first, second, and third), in order to specify the win, place, and show. Thus there are $P(12,3) = 12 \cdot 11 \cdot 10 = 1320$ possibilities.

11. **a)** To specify a bit string of length 10 that contains exactly four 1's, we simply need to choose the four positions that contain the 1's. There are $C(10,4) = 210$ ways to do that.
b) To contain at most four 1's means to contain four 1's, three 1's, two 1's, one 1, or no 1's. Reasoning as in part **(a)**, we see that there are $C(10,4)+C(10,3)+C(10,2)+C(10,1)+C(10,0) = 210+120+45+10+1 = 386$ such strings.
c) To contain at least four 1's means to contain four 1's, five 1's, six 1's, seven 1's, eight 1's, nine 1's, or ten 1's. Reasoning as in part **(b)**, we see that there are $C(10,4)+C(10,5)+C(10,6)+C(10,7)+C(10,8)+C(10,9)+C(10,10) = 210+252+210+120+45+10+1 = 848$ such strings. A simpler approach would be to figure out the number of ways not to have at least four 1's (i.e., to have three 1's, two 1's, one 1, or no 1's) and then subtract that from 2^{10}, the total number of bit strings of length 10. This way we get $1024 - (120+45+10+1) = 848$, fortunately the same answer as before. Solving a combinatorial problem in more than one way is a useful check on the correctness of the answer.
d) To have an equal number of 0's and 1's in this case means to have five 1's. Therefore the answer is $C(10,5) = 252$. Incidentally, this gives us another way to do part **(b)**. If we don't have an equal number of 0's and 1's, then we have either at most four 1's or at least six 1's. By symmetry, having at most four 1's occurs in half of these cases. Therefore the answer to part **(b)** is $(2^{10} - C(10,5))/2 = 386$, as above.

13. We assume that the row has a distinguished head. Consider the order in which the men appear relative to each other. There are n men, and all of the $P(n,n) = n!$ arrangements is allowed. Similarly, there are $n!$ arrangements in which the women can appear. Now the men and women must alternate, and there are the same number of men and women; therefore there are exactly two possibilities: either the row starts with a man and ends with a woman ($MWMW\ldots MW$) or else it starts with a woman and ends with a man ($WMWM\ldots WM$). We have three tasks to perform, then: arrange the men among themselves, arrange the women among themselves, and decide which sex starts the row. By the product rule there are $n! \cdot n! \cdot 2 = 2(n!)^2$ ways in which this can be done.

15. We assume that a combination is called for, not a permutation, since we are told to *select a set*, not *form an arrangement*. We need to choose 5 things from 26, so there are $C(26,5) = 26 \cdot 25 \cdot 24 \cdot 23 \cdot 22/5! = 65{,}780$ ways to do so.

17. We know that there are 2^{100} subsets of a set with 100 elements. All of them have more than two elements except the empty set, the 100 subsets consisting of one element each, and the $C(100,2) = 4950$ subsets with two elements. Therefore the answer is $2^{100} - 5051 \approx 1.3 \times 10^{30}$.

19. **a)** Each flip can be either heads or tails, so there are $2^{10} = 1024$ possible outcomes.

b) To specify an outcome that has exactly two heads, we simply need to choose the two flips that came up heads. There are $C(10, 2) = 45$ such outcomes.

c) To contain at most three tails means to contain three tails, two tails, one tail, or no tails. Reasoning as in part **(b)**, we see that there are $C(10, 3) + C(10, 2) + C(10, 1) + C(10, 0) = 120 + 45 + 10 + 1 = 176$ such outcomes.

d) To have an equal number of heads and tails in this case means to have five heads. Therefore the answer is $C(10, 5) = 252$.

21. **a)** If BCD is to be a substring, then we can think of that block of letters as one superletter, and the problem is to count permutations of five items—the letters A, E, F, and G, and the superletter BCD. Therefore the answer is $P(5, 5) = 5! = 120$.

b) Reasoning as in part **(a)**, we see that the answer is $P(4, 4) = 4! = 24$.

c) As in part **(a)**, we glue BA into one item and glue GF into one item. Therefore we need to permute five items, and there are $P(5, 5) = 5! = 120$ ways to do it.

d) This is similar to part **(c)**. Glue ABC into one item and glue DE into one item, producing four items, so the answer is $P(4, 4) = 4! = 24$.

e) If both ABC and CDE are substrings, then $ABCDE$ has to be a substring. So we are really just permuting three items: $ABCDE$, F, and G. Therefore the answer is $P(3, 3) = 3! = 6$.

f) There are no permutations with both of these substrings, since B cannot be followed by both A and E at the same time.

23. First position the men relative to each other. Since there are eight men, there are $P(8, 8)$ ways to do this. This creates nine slots where a woman (but not more than one woman) may stand: in front of the first man, between the first and second men, ..., between the seventh and eight men, and behind the eighth man. We need to choose five of these positions, in order, for the first through fifth woman to occupy (order matters, because the women are distinct people). This can be done is $P(9, 5)$ ways. Therefore the answer is $P(8, 8) \cdot P(9, 5) = 8! \cdot 9!/4! = 609{,}638{,}400$.

25. **a)** Since the prizes are different, we want an ordered arrangement of four numbers from the set of the first 100 positive integers. Thus there are $P(100, 4) = 94{,}109{,}400$ ways to award the prizes.

b) If the grand prize winner is specified, then we need to choose an ordered set of three tickets to win the other three prizes. This can be done is $P(99, 3) = 941{,}094$ ways.

c) We can first determine which prize the person holding ticket 47 will win (this can be done in 4 ways), and then we can determine the winners of the other three prizes, exactly as in part **(b)**. Therefore the answer is $4P(99, 3) = 3{,}764{,}376$.

d) This is the same calculation as in part **(a)**, except that there are only 99 viable tickets. Therefore the answer is $P(99, 4) = 90{,}345{,}024$. Note that this answer plus the answer to part **(c)** equals the answer to part **(a)**, since the person holding ticket 47 either wins a prize or does not win a prize.

e) This is similar to part **(c)**. There are $4 \cdot 3 = 12$ ways to determine which prizes these two lucky people will win, after which there are $P(98, 2) = 9506$ ways to award the other two prizes. Therefore the answer is $12 \cdot 9506 = 114{,}072$.

f) This is like part **(e)**. There are $P(4, 3) = 24$ ways to choose the prizes for the three people mentioned, and then 97 ways to choose the other winner. This gives $24 \cdot 97 = 2328$ ways in all.

g) Here it is just a matter of ordering the prizes for these four people, so the answer is $P(4, 4) = 24$.

h) This is similar to part **(d)**, except that this time the pool of viable numbers has only 96 numbers in it. Therefore the answer is $P(96, 4) = 79{,}727{,}040$.

i) There are four ways to determine the grand prize winner under these conditions. Then there are $P(99,3)$ ways to award the remaining prizes. This gives an answer of $4P(99,3) = 3{,}764{,}376$.

j) First we need to choose the prizes for the holder of 19 and 47. Since there are four prizes, there are $P(4,2) = 12$ ways to do this. Then there are 96 people who might win the remaining prizes, and there are $P(96,2) = 9120$ ways to award these prizes. Therefore the answer is $12 \cdot 9120 = 109{,}440$.

27. **a)** Since the order of choosing the members is not relevant (the offices are not differentiated), we need to use a combination. The answer is clearly $C(25,4) = 12{,}650$.

b) In contrast, here we need a permutation, since the order matters (we choose first a president, then a vice president, then a secretary, then a treasurer). The answer is clearly $P(25,4) = 303{,}600$.

29. **a)** In this part the permutation $5,6,32,7$, for example, is to be counted, since it contains the consecutive numbers 5, 6, and 7 in their correct order (even though separated by the 32). In order to specify such a 4-permutation, we first need to choose the 3 consecutive integers. They can be anything from $\{1,2,3\}$ to $\{98,99,100\}$; thus there are 98 possibilities. Next we need to decide which slot is to contain a number not in this set; there are 4 possibilities. Finally, we need to decide which of the 97 other positive integers not exceeding 100 is to fill this slot, and there are of course 97 choices. Thus our first attempt at an answer gives us, by the product rule, $98 \cdot 4 \cdot 97$.

Unfortunately, this answer is not correct, because we have counted some 4-permutations more than once. Consider the 4-permutation $4,5,6,7$, for example. We cannot tell whether it arose from choosing 4, 5, and 6 as the consecutive numbers, or from choosing 5, 6, and 7. (These are the only two ways it could have arisen.) In fact, every 4-permutation consisting of 4 consecutive numbers, in order, has been double counted. Therefore to correct our count, we need to subtract the number of such 4-permutations. Clearly there are 97 of them (they can begin with any number from 1 to 97). Further thought shows that every other 4-permutation in our collection arises in a unique way (in other words, there is a unique subsequence of three consecutive integers). Thus our final answer is $98 \cdot 4 \cdot 97 - 97 = 37{,}927$.

b) In this part we are insisting that the consecutive numbers be consecutive in the 4-permutation as well. The analysis in part **(a)** works here, except that there are only 2 places to put the fourth number—in slot 1 or in slot 4. Therefore the answer is $98 \cdot 2 \cdot 97 - 97 = 18{,}915$.

31. We need to be careful here, because strings can have repeated letters.

a) We need to choose the position for the vowel, and this can be done in 6 ways. Next we need to choose the vowel to use, and this can be done in 5 ways. Each of the other five positions in the string can contain any of the 21 consonants, so there are 21^5 ways to fill the rest of the string. Therefore the answer is $6 \cdot 5 \cdot 21^5 = 122{,}523{,}030$.

b) We need to choose the position for the vowels, and this can be done in $C(6,2) = 15$ ways (we need to choose two positions out of six). We need to choose the two vowels (5^2 ways). Each of the other four positions in the string can contain any of the 21 consonants, so there are 21^4 ways to fill the rest of the string. Therefore the answer is $15 \cdot 5^2 \cdot 21^4 = 72{,}930{,}375$.

c) The best way to do this is to count the number of strings with no vowels and subtract this from the total number of strings. We obtain $26^6 - 21^6 = 223{,}149{,}655$.

d) As in part **(c)**, we will do this by subtracting from the total number of strings, the number of strings with no vowels and the number of strings with one vowel (this latter quantity having been computed in part **(a)**). We obtain $26^6 - 21^6 - 6 \cdot 5 \cdot 21^5 = 223149655 - 122523030 = 100{,}626{,}625$.

33. We are told that we must select three of the 10 men and three of the 15 women. This can be done in $C(10,3)C(15,3) = 54{,}600$ ways.

35. To implement the condition that every 0 be immediately followed by a 1, let us think of "gluing" a 1 to the right of each 0. Then the objects we have to work with are eight blocks consisting of the string 01 and two 1's. The question is, then, how many strings are there consisting of these ten objects? This is easy to calculate, for we simply have to choose two of the "positions" in the string to contain the 1's and fill the remaining "positions" with the 01 blocks. Therefore the answer is $C(10,2) = 45$.

37. Perhaps the most straightforward way to do this is to look at the several cases. The string might contain three 1's and seven 0's, four 1's and six 0's, five of each, six 1's and four 0's, or seven 1's and three 0's. In each case we can determine the number of strings by calculating a binomial coefficient, since we simply need to choose the positions for the 1's. Therefore the answer is $C(10,3)+C(10,4)+C(10,5)+C(10,6)+C(10,7) = 120+210+252+210+120 = 912$.

39. To specify such a license plate we need to write down a 3-permutation of the set of 26 letters and follow it by a 3-permutation of the set of 10 digits. By the product rule the answer is therefore $P(26,3) \cdot P(10,3) = 26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 = 11{,}232{,}000$.

41. Designate the head of the table and seat the people clockwise. Clearly there are $P(n,r)$ ways to do this. Because rotations of the table do not make for a "different" seating, this overcounts by a factor of r. Therefore the answer is $P(n,r)/r = n!/(r(n-r)!)$.

43. If there are no ties, then there are $3! = 6$ possible finishes. If two of the horses tie and the third has a different time, then there are 3 ways to decide which horse is not tied and then 2 ways to decide whether that horse finishes first or last. That gives $3 \cdot 2 = 6$ possibilities. Finally, all three horses can tie. So the answer is $6+6+1 = 13$.

45. We can solve this problem by breaking it down into cases depending on the ties. There are four basic cases. (1) If there are unique gold and silver winners, then we can choose these winners in $6 \cdot 5 = 30$ ways. Any nonempty subset of the remaining four runners can win the bronze medal. There are $2^4 - 1 = 15$ ways to choose these people, giving us $30 \cdot 15 = 450$ ways in all for this case. (2) If there is a 2-way tie for first place, then there are $C(6,2) = 15$ ways to choose the gold medalists. Any nonempty subset of the remaining four runners can win the bronze medal, so there are $2^4 - 1 = 15$ ways to choose these people, giving us $15 \cdot 15 = 225$ ways in all for this case. (3) If there is a k-way tie for first with $k \geq 3$, then there are $C(6,k)$ ways to choose the gold medalists (there are no other medals in this case). This gives us $C(6,3)+C(6,4)+C(6,5)+C(6,6) = 20+15+6+1 = 42$ more possibilities. (4) The only other case is that there is a single gold medal winner and a k-way tie for second with $k \geq 2$. We can choose the winner in 6 ways and the silver medalists in $2^5 - C(5,1) - C(5,0) = 32-5-1 = 26$ ways. This gives us $6 \cdot 26 = 156$ possibilities. Putting this all together, the answer is $450+225+42+156 = 873$.

SECTION 6.4 Binomial Coefficients and Identities

In this section we usually write the binomial coefficients using the $\binom{n}{r}$ notation, rather than the $C(n,r)$ notation. These numbers tend to come up in many parts of discrete mathematics.

1. **a)** When $(x+y)^4 = (x+y)(x+y)(x+y)(x+y)$ is expanded, all products of a term in the first sum, a term in the second sum, a term in the third sum, and a term in the fourth sum are added. Terms of the form x^4, x^3y, x^2y^2, xy^3 and y^4 arise. To obtain a term of the form x^4, an x must be chosen in each of the sums, and this can be done in only one way. Thus, the x^4 term in the product has a coefficient of 1. (We can think of this coefficient as $\binom{4}{4}$.) To obtain a term of the form x^3y, an x must be chosen in three of the four sums (and consequently a y in the other sum). Hence, the number of such terms is the number of 3-combinations of four objects, namely $\binom{4}{3} = 4$. Similarly, the number of terms of the form x^2y^2 is the number of ways to pick two of the four sums to obtain x's (and consequently take a y from each of the other two factors). This can be done in $\binom{4}{2} = 6$ ways. By the same reasoning there are $\binom{4}{1} = 4$ ways to obtain the xy^3 terms, and only one way (which we can think of as $\binom{4}{0}$) to obtain a y^4 term. Consequently, the product is $x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$.

b) This is explained in Example 2. The expansion is $\binom{4}{0}x^4 + \binom{4}{1}x^3y + \binom{4}{2}x^2y^2 + \binom{4}{3}xy^3 + \binom{4}{4}y^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$. Note that it does not matter whether we think of the bottom of the binomial coefficient expression as corresponding to the exponent on x, as we did in part **(a)**, or the exponent on y, as we do here.

3. The coefficients are the binomial coefficients $\binom{6}{i}$, as i runs from 0 to 6, namely 1, 6, 15, 20, 15, 6, 1. Therefore $(x+y)^6 = \sum_{i=0}^{6} \binom{6}{i}x^{6-i}y^i = x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6$.

5. There is one term for each i from 0 to 100, so there are 101 terms.

7. By the binomial theorem the term involving x^9 in the expansion of $(2 + (-x))^{19}$ is $\binom{19}{9}2^{10}(-x)^9$. Therefore the coefficient is $\binom{19}{9}2^{10}(-1)^9 = -2^{10}\binom{19}{9} = -94{,}595{,}072$.

9. Using the binomial theorem, we see that the term involving x^{101} in the expansion of $((2x) + (-3y))^{200}$ is $\binom{200}{99}(2x)^{101}(-3y)^{99}$. Therefore the coefficient is $\binom{200}{99}2^{101}(-3)^{99} = -2^{101}3^{99}C(200, 99)$.

11. Let us apply the binomial theorem to the given binomial:

$$(x^2 - x^{-1})^{100} = \sum_{j=0}^{100} \binom{100}{j}(x^2)^{100-j}(-x^{-1})^j$$

$$= \sum_{j=0}^{100} \binom{100}{j}(-1)^j x^{200-2j-j} = \sum_{j=0}^{100} \binom{100}{j}(-1)^j x^{200-3j}$$

Thus the only nonzero coefficients are those of the form $200 - 3j$ where j is an integer between 0 and 100, inclusive, namely 200, 197, 194, ..., 2, −1, −4, ..., −100. If we denote $200 - 3j$ by k, then we have $j = (200-k)/3$. This gives us our answer. The coefficient of x^k is zero for k not in the list just given (namely those vales of k between −100 and 200, inclusive, that are congruent to 2 modulo 3), and for those values of k in the list, the coefficient is $(-1)^{(200-k)/3}\binom{100}{(200-k)/3}$.

13. We are asked simply to display these binomial coefficients. Each can be computed from the formula in Theorem 2 in Section 6.3. Alternatively, we can apply Pascal's identity to the last row of Figure 1(b), adding successive numbers in that row to produce the desired row. We thus obtain

1 9 36 84 126 126 84 36 9 1.

15. There are many ways to see why this is true. By Corollary 1 the sum of *all* the positive numbers $\binom{n}{k}$, as k runs from 0 to n, is 2^n, so certainly each one of them is no bigger than this sum. Another way to see this is to note that $\binom{n}{k}$ counts the number of subsets of an n-set having k elements, and 2^n counts even more—the number of subsets of an n-set with no restriction as to size; so certainly the former is smaller than the latter.

17. We know that

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k(k-1)(k-2)\cdots 2}.$$

Now if we make the numerator of the right-hand side larger by raising each factor up to n, and make the denominator smaller by lowering each factor to 2, then we have certainly not decreased the value, so the left-hand side is less than or equal to this altered expression. But the result is precisely $n^k/2^{k-1}$, as desired.

19. Using the formula (Theorem 2 in Section 6.3) we have

$$\begin{aligned}\binom{n}{k-1} + \binom{n}{k} &= \frac{n!}{(k-1)!(n-(k-1))!} + \frac{n!}{k!(n-k)!} \\ &= \frac{n!k + n!(n-k+1)}{k!(n-k+1)!} \quad \text{(having found a common denominator)} \\ &= \frac{(n+1)n!}{k!((n+1)-k)!} = \frac{(n+1)!}{k!((n+1)-k)!} = \binom{n+1}{k}.\end{aligned}$$

21. **a)** We show that each side counts the number of ways to choose from a set with n elements a subset with k elements and a distinguished element of that set. For the left-hand side, first choose the k-set (this can be done in $\binom{n}{k}$ ways) and then choose one of the k elements in this subset to be the distinguished element (this can be done in k ways). For the right-hand side, first choose the distinguished element out of the entire n-set (this can be done in n ways), and then choose the remaining $k-1$ elements of the subset from the remaining $n-1$ elements of the set (this can be done in $\binom{n-1}{k-1}$ ways).

b) This is straightforward algebra:

$$k\binom{n}{k} = k \cdot \frac{n!}{k!(n-k)!} = \frac{n \cdot (n-1)!}{(k-1)!(n-k)!} = n\binom{n-1}{k-1}.$$

23. This identity can be proved algebraically or combinatorially. Algebraically, we compute as follows, starting with the right-hand side (we use twice the fact that $(x+1)x! = (x+1)!$):

$$\begin{aligned}\frac{(n+1)\binom{n}{k-1}}{k} &= \frac{(n+1)n!}{(k-1)!(n-(k-1))!k} \\ &= \frac{(n+1)!}{k!(n-(k-1))!} \\ &= \frac{(n+1)!}{k!((n+1)-k)!} \\ &= \binom{n+1}{k}\end{aligned}$$

For a combinatorial argument, we need to construct a situation in which both sides count the same thing. Suppose that we have a set of $n+1$ people, and we wish to choose k of them. Clearly there are $\binom{n+1}{k}$ ways to do this. On the other hand, we can choose our set of k people by first choosing one person to be in the set (there are $n+1$ choices), and then choosing $k-1$ additional people to be in the set, from the n people remaining. This can be done in $\binom{n}{k-1}$ ways. Therefore apparently there are $(n+1)\binom{n}{k-1}$ ways to choose the set of k people. However, we have overcounted: there are k ways that every such set can be chosen, since once we have the set, we realize that any of the k people could have been chosen first. Thus we have overcounted by a factor of k, and the real answer is $(n+1)\binom{n}{k-1}/k$ (we correct for the overcounting by dividing by k). Comparing our two approaches, one yielding the answer $\binom{n+1}{k}$, and the other yielding the answer $(n+1)\binom{n}{k-1}/k$, we conclude that $\binom{n+1}{k} = (n+1)\binom{n}{k-1}/k$.

Finally, we are asked to use this identity to give a recursive definition of the $\binom{n}{k}$'s. Note that this identity expresses $\binom{n}{k}$ in terms of $\binom{i}{j}$ for values of i and j less than n and k, respectively (namely $i = n-1$ and $j = k-1$). Thus the identity will be the recursive part of the definition. We need the base cases to handle $n = 0$ or $k = 0$. Our full definition becomes

$$\binom{n}{k} = \begin{cases} 1 & \text{if } k = 0 \\ 0 & \text{if } k > 0 \text{ and } n = 0 \\ n\binom{n-1}{k-1}/k & \text{if } n > 0 \text{ and } k > 0. \end{cases}$$

Actually, if we assume (as we usually do) that $k \leq n$, then we do not need the second line of the definition. Note that $\binom{n}{k} = 0$ for $n < k$ under the definition given here, which is consistent with the combinatorial definition, since there are no ways to choose k different elements from a set with fewer than k elements.

25. We use Pascal's identity twice (Theorem 2 of this section) and Corollary 1 of the previous section:

$$\begin{aligned}\binom{2n}{n+1} + \binom{2n}{n} = \binom{2n+1}{n+1} &= \frac{1}{2}\left(\binom{2n+1}{n+1} + \binom{2n+1}{n+1}\right) \\ &= \frac{1}{2}\left(\binom{2n+1}{n+1} + \binom{2n+1}{n}\right) = \frac{1}{2}\left(\binom{2n+2}{n+1}\right)\end{aligned}$$

27. The reason this is called the "hockey stick identity" is that the binomial coefficients being summed lie along a line in Pascal's triangle, and the sum is the entry in the triangle that, together with that line, forms a shape vaguely resembling a hockey stick.

a) We need to find something to count so that the left-hand side of the equation counts it in one way and the right-hand side counts it in a different way. After much thought, we might try the following. We will count the number of bit strings of length $n+r+1$ containing exactly r 0's and $n+1$ 1's. There are $\binom{n+r+1}{r}$ such strings, since a string is completely specified by deciding which r of the bits are to be the 0's. To see that the left-hand side of the identity counts the same thing, let $l+1$ be the position of the last 1 in the string. Since there are $n+1$ 1's, we know that l cannot be less than n. Thus there are disjoint cases for each l from n to $n+r$. For each such l, we completely determine the string by deciding which of the l positions in the string before the last 1 are to be 0's. Since there are n 1's in this range, there are $l-n$ 0's. Thus there are $\binom{l}{l-n}$ ways to choose the positions of the 0's. Now by the sum rule the total number of bit strings will be $\sum_{l=n}^{n+r}\binom{l}{l-n}$. By making the change of variable $k = l-n$, this transforms into the left-hand side, and we are finished.

b) We need to prove this by induction on r; Pascal's identity will enter at the crucial step. We let $P(r)$ be the statement to be proved. The basis step is clear, since the equation reduces to $\binom{n}{0} = \binom{n+1}{0}$, which is the true proposition $1 = 1$. Assuming the inductive hypothesis, we derive $P(r+1)$ in the usual way:

$$\begin{aligned}\sum_{k=0}^{r+1}\binom{n+k}{k} &= \left(\sum_{k=0}^{r}\binom{n+k}{k}\right) + \binom{n+r+1}{r+1} \\ &= \binom{n+r+1}{r} + \binom{n+r+1}{r+1} \quad \text{(by the inductive hypothesis)} \\ &= \binom{n+(r+1)+1}{r+1} \quad \text{(by Pascal's identity)}\end{aligned}$$

29. We will follow the hint and count the number of ways to choose a committee with leader from a set of n people. Note that the size of the committee is not specified, although it clearly needs to have at least one person (its leader). On the one hand, we can choose the leader first, in any of n ways. We can then choose the rest of the committee, which can be any subset of the remaining $n-1$ people; this can be done in 2^{n-1} ways since there are this many subsets. Therefore the right-hand side of the proposed identity counts this.

On the other hand, we can organize our count by the size of the committee. Let k be the number of people who will serve on the committee. The number of ways to select a committee with k people is clearly $\binom{n}{k}$, and once we have chosen the committee, there are clearly k ways in which to choose its leader. By the sum rule the left-hand side of the proposed identity therefore also counts the number of such committees. Since the two sides count the same quantity, they must be equal.

31. Corollary 2 says that $\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots \pm \binom{n}{n} = 0$. If we put all the negative terms on the other side, we obtain $\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = \binom{n}{1} + \binom{n}{3} + \cdots$ (one side ends at $\binom{n}{n}$ and the other side ends at $\binom{n}{n-1}$—which is which depends on whether n is even or odd). Now the left-hand side counts the number of subsets with even cardinality of a set with n elements, and the right-hand side counts the number of subsets with odd cardinality of the same set. That these two quantities are equal is precisely what we wanted to prove.

33. **a)** Clearly a path of the desired type must consist of m moves to the right and n moves up. Therefore each such path can be represented by a bit string consisting of m 0's and n 1's, with the 0's representing moves to the right and the 1's representing moves up. Note that the total length of this bit string is $m + n$.

b) We know from this section that the number of bit strings of length $m + n$ containing exactly n 1's is $\binom{m+n}{n}$, since one need only specify the positions of the 1's. Note that this is the same as $\binom{m+n}{m}$.

35. We saw in Exercise 33 that the number of paths of length n was the same as the number of bit strings of length n, which we know to be 2^n, the right-hand side of the identity. On the other hand, a path of length n must end up at some point the sum of whose coordinates is n, say at $(n-k, k)$ for some k from 0 to n. We saw in Exercise 33 that the number of paths ending up at $(n-k, k)$ was equal to $\binom{n-k+k}{k} = \binom{n}{k}$. Therefore the left-hand side of the identity counts the number of such paths, too.

37. The right-hand side of the identity we are asked to prove counts, by Exercise 33, the number of paths from $(0, 0)$ to $(n+1, r)$. Now let us count these paths by breaking them down into $r+1$ cases, depending on how many steps upward they begin with. Let k be the number of steps upward they begin with before taking a step to the right. Then k can take any value from 0 to r. The number of paths from $(0, 0)$ to $(n+1, r)$ that begin with exactly k steps upward before turning to the right is clearly the same as the number of paths from $(1, k)$ to $(n+1, r)$, since after these k upward steps and the move to the right we have reached $(1, k)$. This latter quantity is the same as the number of paths from $(0, 0)$ to $(n+1-1, r-k) = (n, r-k)$, since we can relabel our diagram to make $(1, k)$ the origin. From Exercise 33, this latter quantity is $\binom{n+r-k}{r-k}$. Therefore the total number of paths is the desired sum

$$\sum_{k=0}^{r} \binom{n+r-k}{r-k} = \sum_{k=0}^{r} \binom{n+k}{k},$$

where the equality comes from changing the dummy variable from k to $r-k$. Since both sides count the same thing, they are equal.

39. **a)** This looks like the third negatively sloping diagonal of Pascal's triangle, starting with the leftmost entry in the second row and reading down and to the right. In other words, the nth term of this sequence is $\binom{n+1}{n-1} = \binom{n+1}{2}$.

b) This looks like the fourth negatively sloping diagonal of Pascal's triangle, starting with the leftmost entry in the third row and reading down and to the right. In other words, the nth term of this sequence is $\binom{n+2}{n-1} = \binom{n+2}{3}$.

c) These seem to be the entries reading straight down the middle of the Pascal's triangle. Only every other row has a middle element. The first entry in the sequence is $\binom{0}{0}$, the second is $\binom{2}{1}$, the third is $\binom{4}{2}$, the fourth is $\binom{6}{3}$, and so on. In general, then, the nth term is $\binom{2n-2}{n-1}$.

d) These seem to be the entries reading down the middle of the Pascal's triangle. Only every other row has an exact middle element, but in the other rows, there are two elements sharing the middle. The first entry in the sequence is $\binom{0}{0}$, the second is $\binom{1}{0}$, the third is $\binom{2}{1}$, the fourth is $\binom{3}{1}$, the fifth is $\binom{4}{2}$, the fourth is $\binom{5}{2}$, and so on. In general, then, the nth term is $\binom{n-1}{\lfloor (n-1)/2 \rfloor}$.

e) One pattern here is a wandering through Pascal's triangle according to the following rule. The nth term is in the nth row of the triangle. We increase the position in that row as we go down, but as soon as we are about to reach the middle of the row, we jump back to the start of the next row. For example, the fifth term is the first entry in row 5; the sixth term is the second entry in row 6; the seventh term is the third entry in row 7; the eighth term is the fourth entry in row 8. The next entry following that pattern would take us to the middle of the ninth row, so instead we jump back to the beginning, and the ninth term is the first entry of row 9. To come up with a formula here, we see that the nth entry is $\binom{n-1}{k-1}$ for a particular k; let us determine k as a function of n. A little playing around with the pattern reveals that k is n minus the largest power of 2 less than n (where for this purpose we consider 0 to be the largest power of 2 less than 1). For example, the 14th term has $k = 14 - 8 = 6$, so it is $\binom{13}{5} = 1287$.

f) The terms seem to come from every third row, so the nth term is $\binom{3n-3}{k}$ for some k. A little observation indicates that in fact these terms are $\binom{0}{0}$, $\binom{3}{1}$, $\binom{6}{2}$, $\binom{9}{3}$, $\binom{12}{4}$, and so on. Thus the nth term is $\binom{3n-3}{n-1}$.

SECTION 6.5 Generalized Permutations and Combinations

As in Section 6.3, we have formulae that give us the answers to some combinatorial problems, if we can figure out which formula applies to which problem, and in what way it applies. Here, even more than in previous sections, the ability to see a problem from the right perspective is the key to solving it. Expect to spend several minutes staring at each problem before any insight comes. Reread the examples in the section and try to imagine yourself going through the thought processes explained there. Gradually your mind will begin to think in the same terms. In particular, ask yourself what is being selected from what, whether ordered or unordered selections are to be made, and whether repetition is allowed. In most cases, after you have answered these questions, you can find the appropriate formula from Table 1.

1. Since order is important here, and since repetition is allowed, this is a simple application of the product rule. There are 3 ways in which the first element can be selected, 3 ways in which the second element can be selected, and so on, with finally 3 ways in which the fifth element can be selected, so there are $3^5 = 243$ ways in which the 5 elements can be selected. The general formula is that there are n^k ways to select k elements from a set of n elements, in order, with unlimited repetition allowed.

3. Since we are considering strings, clearly order matters. The choice for each position in the string is from the set of 26 letters. Therefore, using the same reasoning as in Exercise 1, we see that there are $26^6 = 308{,}915{,}776$ strings.

5. We assume that the jobs and the employees are distinguishable. For each job, we have to decide which employee gets that job. Thus there are 5 ways in which the first job can be assigned, 5 ways in which the second job can be assigned, and 5 ways in which the third job can be assigned. Therefore, by the multiplication principle (just as in Exercise 1) there are $5^3 = 125$ ways in which the assignments can be made. (Note that we do not require that every employee get at least one job.)

7. Since the selection is to be an unordered one, Theorem 2 applies. We want to choose $r = 3$ items from a set with $n = 5$ elements. Theorem 2 tells us that there are $C(5 + 3 - 1, 3) = C(7, 3) = 7 \cdot 6 \cdot 5/(3 \cdot 2) = 35$ ways to do so. (Equivalently, this problem is asking us to count the number of nonnegative integer solutions to $x_1 + x_2 + x_3 + x_4 + x_5 = 3$, where x_i represents the number of times that the ith element of the 5-element set gets selected.)